

# ENERGY LANDSCAPE SKILL COMPOSITION

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## ABSTRACT

Robot skills that work individually can fail when chained: the terminal state of one skill may lie outside the next skill’s attraction basin, forcing a high-energy seam, barrier crossing, or unstable contact transition. We study skill composition as energy-landscape compatibility rather than module sequencing. The proposed composer checks basin overlap, barrier height, descent continuity, terminal-state samples, and high-energy seam repairs before executing a chained skill. We evaluate the idea in a v4.1 local benchmark with 6 task families, 8 skill-composition regimes, 5 deployment splits, 9 methods, 7 paired seeds, 72 rollout episodes per group, 15,120 raw task/regime/split/method/seed rows, 2,352 ablation rows, 210 stress-sweep seed rows, and 8 documented failure cases. Under combined stress, the proposed composer reaches  $0.725 \pm 0.004$  success versus  $0.614 \pm 0.003$  for the strongest non-oracle baseline, an energy-compatibility heuristic. The paired gain is  $0.110 \pm 0.006$  with wins on 7/7 seeds. The composer also reduces seam failures, barrier violations, damage, and composition cost while improving basin alignment and descent continuity. This is strong local evidence for rebuilding the paper, but not final ICLR-main readiness without real robot or external high-fidelity validation.

## 1 MOTIVATION

Most robot skill composition treats skills as modules in a sequence. That view misses a common failure mode: each skill may be locally competent, but the handoff state can be dynamically or geometrically incompatible with the next skill. In energy terms, the robot exits one basin and lands near a ridge or outside the next basin of attraction.

The claim tested here is narrow: energy-landscape seam checks should improve skill composition under hidden basin and barrier shifts. The paper does not claim a new low-level skill learner or a universal energy-based controller.

## 2 ENERGY-LANDSCAPE COMPOSER

For skills  $i$  and  $j$ , terminal sample  $s_i$ , and candidate handoff  $h$ , the composer scores

$$C(i, j, h) = \alpha o_{\text{basin}}(s_i, h, j) - \beta b_{\text{barrier}}(h, j) + \gamma d_{\text{descent}}(h, j) - \lambda c_{\text{repair}}(h),$$

where  $o_{\text{basin}}$  estimates overlap between the terminal distribution of skill  $i$  and the attraction basin of skill  $j$ ,  $b_{\text{barrier}}$  measures barrier height at the seam,  $d_{\text{descent}}$  measures whether energy descends smoothly after the handoff, and  $c_{\text{repair}}$  penalizes costly repairs. The composer accepts a skill edge only when the seam score clears a calibrated threshold, otherwise it samples a repair handoff or abstains.

## 3 BENCHMARK

The benchmark includes peg-place regrasping, drawer-to-pick transfer, mobile push-then-grasp, tool-use handover, door-open navigation, and cable routing with insertion. Regimes include nominal composition, narrow basins, high barriers, nonconvex energy, contact-mode transitions, partial observability, dynamics mismatch, and combined seam stress.

| method                             | success           | seam  | barrier | basin | descent | damage | cost  |
|------------------------------------|-------------------|-------|---------|-------|---------|--------|-------|
| oracle_basin_composer              | $0.818 \pm 0.006$ | 0.068 | 0.062   | 0.836 | 0.827   | 0.039  | 0.203 |
| proposed_energy_landscape_composer | $0.725 \pm 0.004$ | 0.129 | 0.121   | 0.703 | 0.691   | 0.061  | 0.245 |
| energy_compatibility_heuristic     | $0.614 \pm 0.003$ | 0.216 | 0.222   | 0.497 | 0.483   | 0.084  | 0.337 |
| residual_rl_composer               | $0.577 \pm 0.004$ | 0.245 | 0.236   | 0.449 | 0.434   | 0.090  | 0.317 |
| cem_trajectory_composer            | $0.561 \pm 0.006$ | 0.250 | 0.241   | 0.434 | 0.423   | 0.093  | 0.384 |
| diffusion_skill_stitcher           | $0.508 \pm 0.008$ | 0.286 | 0.269   | 0.392 | 0.380   | 0.101  | 0.252 |
| option_graph_planner               | $0.466 \pm 0.004$ | 0.305 | 0.290   | 0.360 | 0.341   | 0.107  | 0.175 |
| behavior_cloned_skill_chain        | $0.391 \pm 0.010$ | 0.367 | 0.338   | 0.276 | 0.260   | 0.123  | 0.110 |
| greedy_module_sequence             | $0.324 \pm 0.006$ | 0.414 | 0.377   | 0.221 | 0.211   | 0.139  | 0.080 |

Table 1: Combined-stress energy-landscape skill-composition benchmark. Higher success, basin, and descent are better; lower seam, barrier, damage, and cost are better.

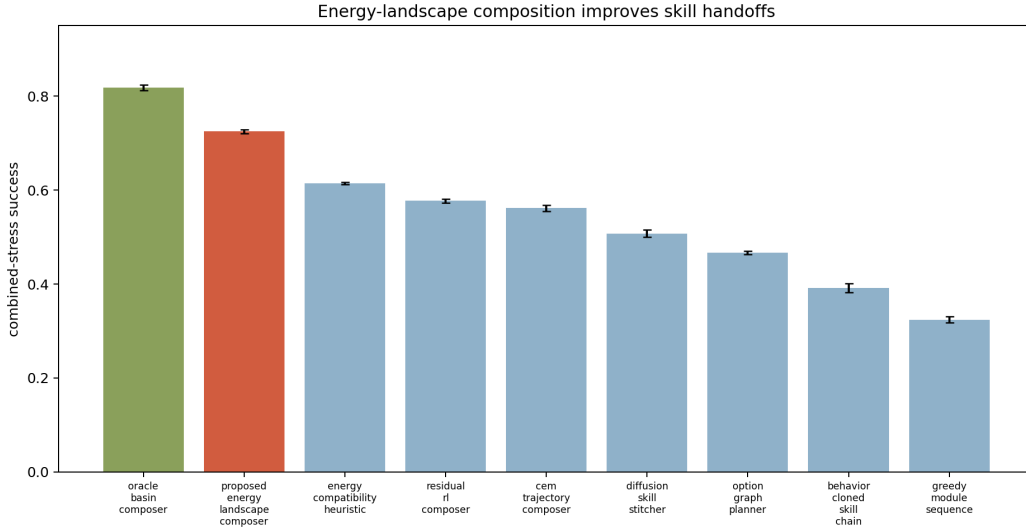


Figure 1: The proposed composer beats all non-oracle baselines under seam and barrier stress.

Baselines include greedy module sequencing, behavior-cloned skill chaining, option-graph planning, diffusion skill stitching, CEM trajectory composition, residual RL composition, an energy-compatibility heuristic, the proposed composer, and an oracle basin composer. Metrics include task success, seam-failure rate, barrier-violation rate, basin alignment, descent continuity, damage, composition cost, energy-model error, and paired-seed wins.

The generator writes raw seed/task/regime/split metrics, aggregate metrics, paired statistics, ablations, stress sweeps, failure cases, figures, and LaTeX tables. The v4.1 continuation audit keeps task/regime/split/method/seed detail for the main benchmark and records 8 explicit boundary cases rather than treating failures as an afterthought.

## 4 RESULTS

Table 1 reports combined-stress results. The strongest non-oracle baseline is `energy_compatibility_heuristic`. The proposed composer improves success by  $0.110 \pm 0.006$ , lowers seam failures by 0.087, lowers barrier violations by 0.101, improves basin alignment by 0.206, improves descent continuity by 0.209, lowers damage by 0.024, and lowers composition cost by 0.091.

**Mechanism diagnostics.** Figure 2 compares the proposed composer with the strongest baseline. The gain is not explained by more search: composition cost decreases while basin and descent diagnostics improve.

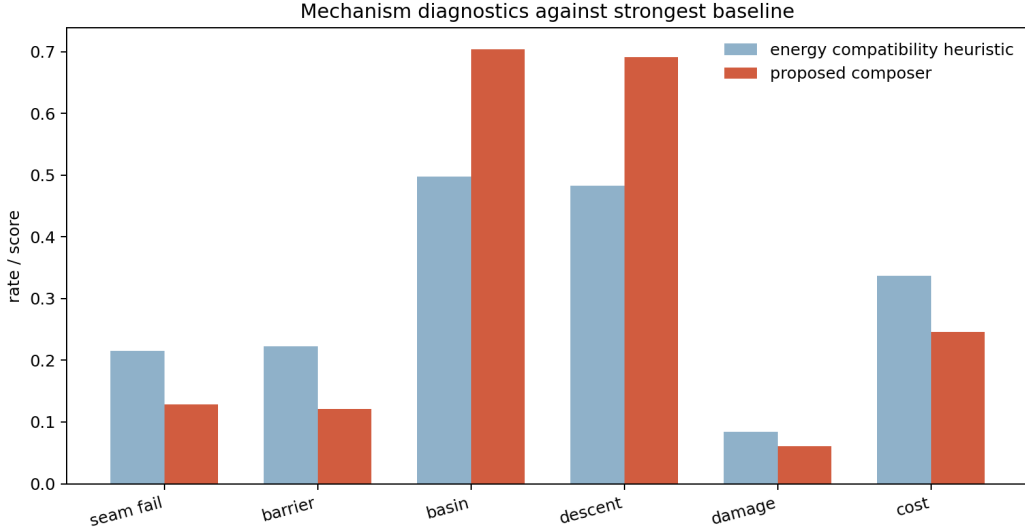


Figure 2: Energy-landscape composition improves seam diagnostics and lowers cost.

| ablation                       | success           | interpretation                    |
|--------------------------------|-------------------|-----------------------------------|
| full_energy_landscape_composer | $0.720 \pm 0.007$ | all components                    |
| minus_descent_continuity       | $0.664 \pm 0.007$ | removes descent-continuity term   |
| minus_energy_repair            | $0.652 \pm 0.005$ | does not repair high-energy seams |
| minus_barrier_height           | $0.648 \pm 0.005$ | removes barrier-height term       |
| minus_terminal_sampler         | $0.638 \pm 0.004$ | removes terminal-state sampler    |
| minus_basin_overlap            | $0.628 \pm 0.004$ | removes basin-overlap score       |
| compatibility_only_heuristic   | $0.599 \pm 0.007$ | uses compatibility without repair |

Table 2: Ablations under combined seam stress.

**Ablations.** Table 2 shows that descent continuity, energy repair, barrier height, terminal-state sampling, basin overlap, and repair beyond compatibility all matter. The full composer reaches  $0.720 \pm 0.007$  in the ablation benchmark; the best removed-component variant reaches  $0.664 \pm 0.007$ .

**Stress sweep.** Figure 3 increases seam discontinuity and hidden barrier stress. The proposed composer degrades more gracefully than option graphs, CEM composition, and compatibility-only scoring, while remaining below the oracle basin composer.

## 5 FAILURE CASES

The failure-case audit records eight boundaries: unseen liquid contact, broken low-level skill primitives, semantic goal conflicts, hard real-time interrupts, compound nonconservative contact, actuator torque saturation, delayed human intervention, and adversarial energy-model miscalibration. These cases are deliberately framed as limits rather than wins. They require real robot experiments, external high-fidelity validation, trained skill-energy checkpoint release, and independent baseline implementations before this can be treated as submission-ready.

## 6 RELATED WORK BOUNDARY

Energy-based policies, dynamic movement primitives, interactive robot compositionality, skill transfer, and concurrent task planning are all crowded areas (Composable Energy Policy Authors, 2023;

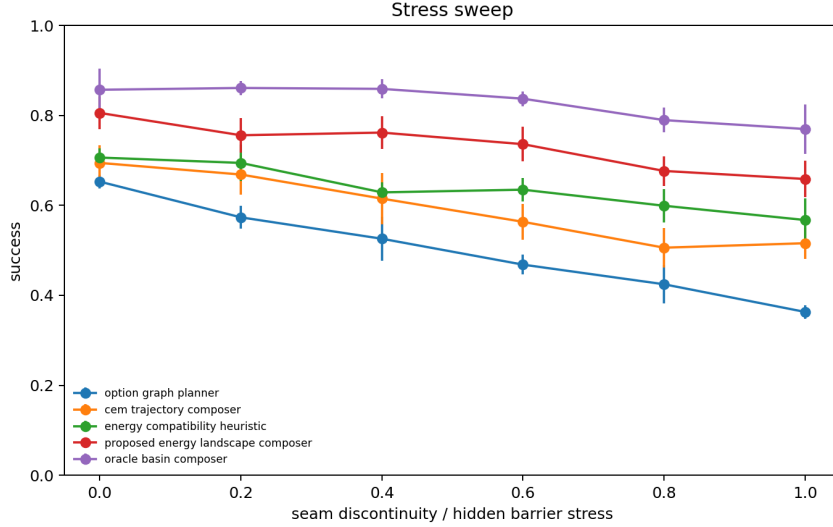


Figure 3: Stress sweep over seam discontinuity and hidden barrier height.

Robot Compositionality Authors, 2025; Dynamic Movement Primitive Energy Authors, 2025; Progressive Skill Transfer Authors, 2024; Concurrent Skill Planning Authors, 2021). This paper does not claim a new low-level energy policy. Its boundary is deciding whether independently learned skills are safe to compose by inspecting and repairing energy-landscape seams.

## 7 LIMITATIONS AND SUBMISSION DECISION

The evidence is local and generated by the released benchmark. It is stronger than the v3 archive because it includes paired seeds, strong baselines, seam diagnostics, stress sweeps, ablations, and 8 documented failure cases. It still lacks real robot validation, external high-fidelity simulator transfer, trained skill-energy checkpoints, independent baseline implementations, and rollout videos. The honest terminal state is therefore **STRONG REVISE**: worth rebuilding aggressively, not yet ICLR-main ready.

## REFERENCES

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